ASSIGNMENT 8

Title: Write an Embedded C program for temperature sensor interfacing using ADC and display on LCD

Objective : To learn interfacing of temperature sensor with PIC18FXXX micro controller

Problem Statement : Write an Embedded C program for temperature sensor interfacing using ADC and display on LCD

Input:

```
#include <pic18f4550.h>
#include <stdio.h>

#define LCD_EN LATAbits.LA1
#define LCD_RS LATAbits.LA0
#define LCDPORT LATB

unsigned char str[16];


void lcd_delay(unsigned int time)
{
       unsigned int i , j ;

    for(i = 0; i < time; i++)
    {
       for(j=0;j<100;j++);
```

```
       }
}



void SendInstruction(unsigned char command)
{
     LCD_RS = 0;                                    // RS low : Instruction
     LCDPORT = command;
     LCD_EN = 1;                                    // EN High
     lcd_delay(10);
     LCD_EN = 0;                                    // EN Low; command sampled at
EN falling edge
     lcd_delay(10);
}

void SendData(unsigned char lcddata)
{
     LCD_RS = 1;          // RS HIGH : DATA
     LCDPORT = lcddata;
     LCD_EN = 1;          // EN High
     lcd_delay(10);
     LCD_EN = 0;          // EN Low; data sampled at EN falling edge
     lcd_delay(10);
}

void InitLCD(void)
{
    ADCON1 = 0x0F;
    TRISB = 0x00; //set data port as output
    TRISAbits.RA0 = 0; //RS pin
    TRISAbits.RA1 = 0; // EN pin

    SendInstruction(0x38);      //8 bit mode, 2 line,5x7 dots
    SendInstruction(0x06);//entry mode
```

```
    SendInstruction(0x0C);//Display ON cursor OFF
    SendInstruction(0x01);      //Clear display
    SendInstruction(0x80);      //set address to 0
}

void LCD_display(unsigned int row, unsigned int pos, unsigned char *ch)
{
    if(row==1)
        SendInstruction(0x80 | (pos-1));
    else
        SendInstruction(0xC0 | (pos-1));

    while(*ch)
        SendData(*ch++);
}

void ADCInit(void)
{
    TRISEbits.RE2 = 1;                  //ADC channel 7 input

    ADCON1 = 0b00000111;                //Ref voltages Vdd & Vss; AN0 - AN7
channels Analog
    ADCON2 = 0b10101110;                //Right justified; Acquisition time 4T;
Conversion clock Fosc/64
}

unsigned short Read_Temp(void)
{
    ADCON0 = 0b00011101;      //ADC on; Select channel;
    GODONE = 1;              //Start Conversion

    while(GO_DONE == 1 );    //Wait till A/D conversion is complete
    return ADRES;                       //Return ADC result
}
```

```c
int main(void)
{
    unsigned int temp;
    InitLCD();
    ADCInit();
    LCD_display(1,1,"Temperature:");
    while(1)
    {
        temp = Read_Temp();
        temp = ((temp * 500) / 1023);
        sprintf(str,"%d'C  ",temp);
        LCD_display(2,1,str);
        lcd_delay(9000);
    }
    return 0;
}
```